

Models and Theories. What are they?

A Model

A *model* is a representation ('re-presentation') of some phenomena. It may be empirical, or based on physical laws, laboratory or numerical, analog or digital.

Examples

- Rotating tank in the laboratory.
- Computer model of a virtual reality (the video game SimCity, the Holodeck from Star Trek, the Matrix movie).
- Conceptual/mathematical models (e.g., the Ising model, the Big Bang and Steady State models in cosmology).
- Complicated numerical models (e.g., GCM)

Properties

- Comprehensiveness and verisimilitude are often virtues.
- Predictive power is a virtue
- Parsimony is not a requirement.
- Generality is not a requirement.
- A priori nature not a requirement.